

```

import pandas as pd
import numpy as np
import re
import string

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import confusion_matrix, classification_report

import seaborn as sns
import matplotlib.pyplot as plt

```

```

# Load the uploaded CSV
df = pd.read_csv('/content/apple_iphone_11_reviews.csv')

# Show first 5 rows
df.head()

```

	index	product	helpful_count	total_comments	url	review_country	reviewed_at	review_text
0	0	Apple iPhone XR (64GB) - Black	5,087 people found this helpful	24	https://www.amazon.in/Apple-iPhone-XR-64GB-Bla...	India	2018-12-12	
1	1	Apple iPhone XR (64GB) - Black	2,822 people found this helpful	6	https://www.amazon.in/Apple-iPhone-XR-64GB-Bla...	India	2018-11-17	V exp y iPh P A pho e comi
2	2	Apple iPhone XR (64GB) - Black	1,798 people found this helpful	0	https://www.amazon.in/Apple-iPhone-XR-64GB-Bla...	India	2019-01-27	So iPh jus pi
3	3	Apple iPhone XR (64GB) - Black	1,366 people found this helpful	14	https://www.amazon.in/Apple-iPhone-XR-64GB-Bla...	India	2019-05-02	I've l andr al
4	4	Apple iPhone XR (64GB) - Black	536 people found this helpful	5	https://www.amazon.in/Apple-iPhone-XR-64GB-Bla...	India	2019-05-24	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```

print("Shape:", df.shape)
print("\nColumns:", df.columns)

```

Shape: (5010, 12)

Columns: Index(['index', 'product', 'helpful_count', 'total_comments', 'url', 'review_country', 'reviewed_at', 'review_text', 'review_rating',

```
'product_company', 'profile_name', 'review_title'],
dtype='object')
```

```
df = df[['review_text', 'review_rating']]

df.head()
```

	review_text	review_rating	
0	NOTE:	3.0 out of 5 stars	
1	Very bad experience with this iPhone xr phone....	1.0 out of 5 stars	
2	Amazing phone with amazing camera coming from ...	5.0 out of 5 stars	
3	So I got the iPhone XR just today. The product...	1.0 out of 5 stars	
4	I've been an android user all my life until I ...	5.0 out of 5 stars	

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```
df['sentiment'] = df['review_rating'].apply(lambda x: 'Positive' if float(x.split(' ')[0]) >= 4 else 'Negati

df['sentiment'].value_counts()
```

	count
sentiment	
Positive	4451
Negative	559

dtype: int64

```
def clean_text(text):
    # Ensure text is a string
    text = str(text)

    # Lowercase
    text = text.lower()

    # Remove punctuation
    text = text.translate(str.maketrans('', '', string.punctuation))

    # Remove numbers
    text = re.sub(r'\d+', '', text)

    # Remove stopwords
    words = text.split()
    words = [w for w in words if w not in ENGLISH_STOP_WORDS]

    return " ".join(words)

df['cleaned_review'] = df['review_text'].apply(clean_text)

df[['review_text', 'cleaned_review']].head()
```

	review_text	cleaned_review	
0	NOTE:	note	
1	Very bad experience with this iPhone xr phone....	bad experience iphone xr phone camera defectiv...	
2	Amazing phone with amazing camera coming from ...	amazing phone amazing camera coming iphone plu...	
3	So I got the iPhone XR just today. The product...	got iphone xr just today product looks amazing...	
4	I've been an android user all my life until I ...	ive android user life decided try iphone xr io...	

```
vectorizer = TfidfVectorizer(max_features=5000)
```

```
X = vectorizer.fit_transform(df['cleaned_review'])
y = df['sentiment']
```

```
print("Feature matrix shape:", X.shape)
print("Sample feature names:", vectorizer.get_feature_names_out()[:10])
```

```
Feature matrix shape: (5010, 3855)
Sample feature names: ['aaple' 'abilities' 'ability' 'able' 'abruptly' 'absence' 'absolute'
'absolutely' 'abt' 'ac']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
print("Train:", X_train.shape)
print("Test:", X_test.shape)
```

```
Train: (4008, 3855)
Test: (1002, 3855)
```

```
nb_model = MultinomialNB()
```

```
nb_model.fit(X_train, y_train)
```

```
print("Model trained successfully!")
```

```
Model trained successfully!
```

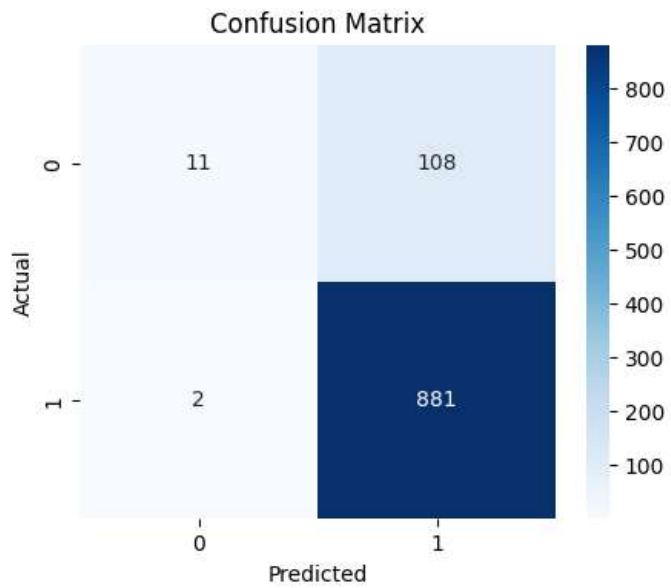
```
y_pred = nb_model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, average='weighted'))
print("Recall:", recall_score(y_test, y_pred, average='weighted'))
print("F1 Score:", f1_score(y_test, y_pred, average='weighted'))
```

```
Accuracy: 0.8902195608782435
Precision: 0.885496643021028
Recall: 0.8902195608782435
F1 Score: 0.8492491512701095
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
Negative	0.85	0.09	0.17	119
Positive	0.89	1.00	0.94	883
accuracy			0.89	1002
macro avg	0.87	0.55	0.55	1002
weighted avg	0.89	0.89	0.85	1002